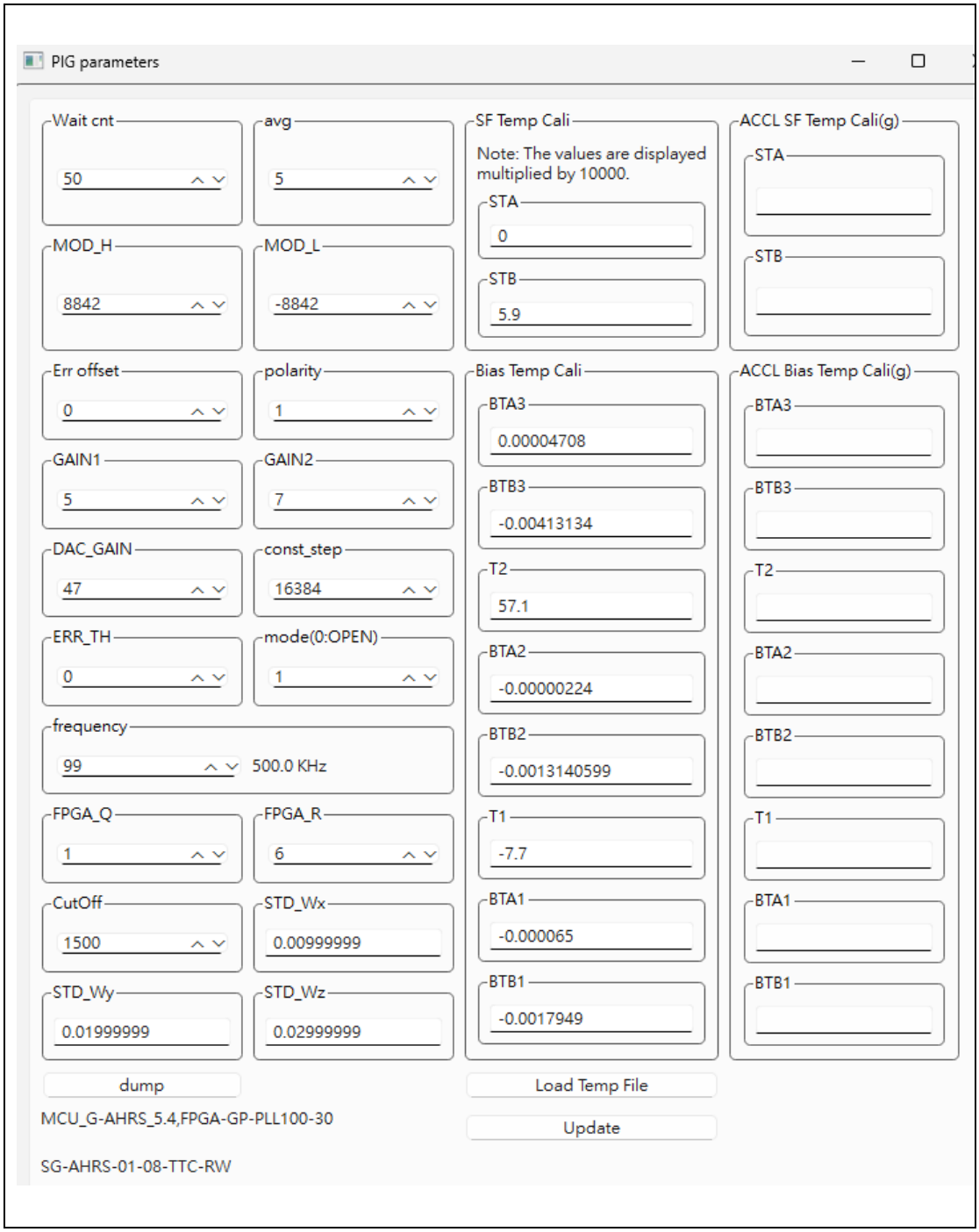
TODO:

1. 量測 SYNC，FPGA TX，MCU TX，紀錄時間 => OK
2. 增加 PD_TEMP 精度 => OK
3. MEMS IMU BW 設死在 25 Hz，增加 lib 功能 => OK
4. FOG 轉速之數位濾波，若 SYNC 是 200 Hz，那 BW 只能是 100 嗎? => OK，加在 MCU

2026-03-11 開始

- 使用小黑作驗證
- 目前可以用 GAHRS-01-02 GUI 運作
- MCU code 路徑:
  - https://drive.google.com/open?id=1Z4sZbMl9m8yL76bm1i8ne437Z-4hXzsw&usp=drive_fs
- 舊版 FPGA 程式使用:
  - FPGA-GP-PLL100-30
- 目標: 根據 HINS 經驗更新 FPGA code，將浮點數運算移到 MCU 來降低 FPGA 計算時間
- 需支援 MCU 韌體: GHARS_NCU_G-AHRS_5.21

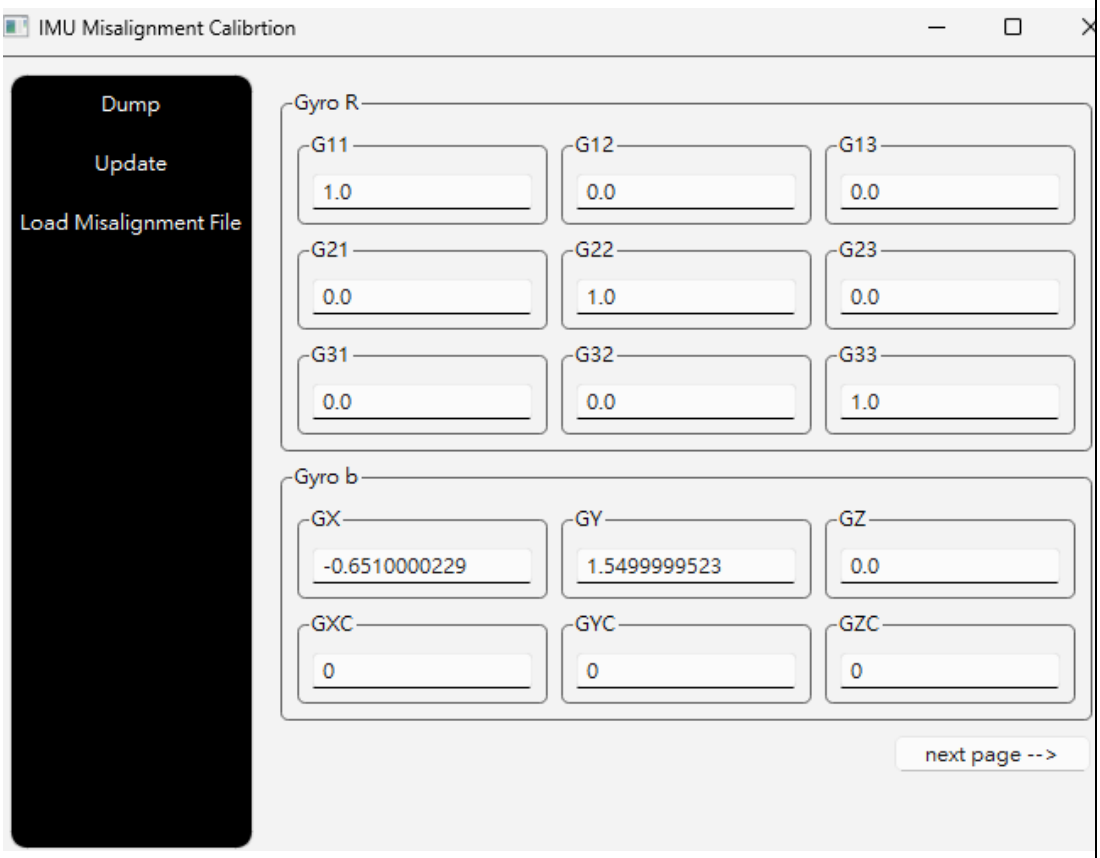
更新前參數:

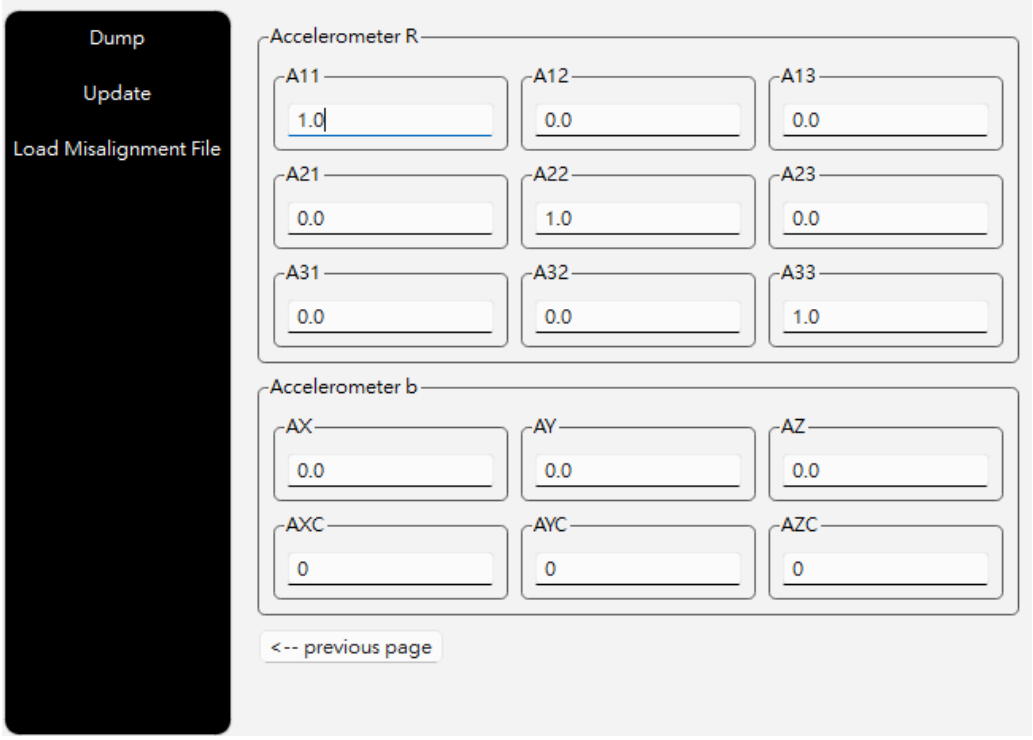

PIG parameters
Wait cnt
50
avg
5
SF Temp Cali
Note: The values are displayed multiplied by 10000.
STA
0
STB
5.9
ACCL SF Temp Cali(g)
STA
STB
MOD_H
8842
MOD_L
-8842
Err offset
0
polarity
1
Bias Temp Cali
BTA3
0.00004708
BTB3
-0.00413134
T2
57.1
BTA2
-0.00000224
BTB2
-0.0013140599
T1
-7.7
BTA1
-0.000065
BTB1
-0.0017949
ACCL Bias Temp Cali(g)
BTA3
BTB3
T2
BTA2
BTB2
T1
BTA1
BTB1
GAIN1
5
GAIN2
7
DAC_GAIN
47
const_step
16384
ERR_TH
0
mode(0:OPEN)
1
frequency
99
500.0 KHz
FPGA_Q
1
FPGA_R
6
CutOff
1500
STD_Wx
0.00999999
STD_Wy
0.01999999
STD_Wz
0.02999999
dump
Load Temp File
Update
MCU_G-AHRS_5.4,FPGA-GP-PLL100-30
SG-AHRS-01-08-TTC-RW

新程式找參數:

STEP 0: 4.39v

STEP 16384:2.21


IMU Misalignment Calibrtion
Dump
Update
Load Misalignment File
Gyro R
G11
1.0
G12
0.0
G13
0.0
G21
0.0
G22
1.0
G23
0.0
G31
0.0
G32
0.0
G33
1.0
Gyro b
GX
-0.6510000229
GY
1.5499999523
GZ
0.0
GXC
0
GYC
0
GZC
0
next page -->


Dump
Update
Load Misalignment File
Accelerometer R
A11
1.0
A12
0.0
A13
0.0
A21
0.0
A22
1.0
A23
0.0
A31
0.0
A32
0.0
A33
1.0
Accelerometer b
AX
0.0
AY
0.0
AZ
0.0
AXC
0
AYC
0
AZC
0
<-- previous page

CMD 行為:

GUI 按下 connect ，按下 stop:

```
cmd, value, ch: 2, 4, 2
```

按下 read:

16: CMD_FOG_TIMER_RST

```
cmd, value, ch: 16, 1, 3
cmd, value, ch: 2, 2, 2
```

按下參數 dump:

```
cmd, value, ch: 102, 5, 2
{"FREQ":10, "MOD_H":8842, "MOD_L":-8842, "ERR_OFFSET":0, "POLARITY":1, "WAIT_CNT":0, "ERR_TH":0, "ERR_AVG":0, "GAIN1":5, "GAIN
cmd, value, ch: 9, 8842, 2
cmd, value, ch: 10, -8842, 2
cmd, value, ch: 12, 1, 2
cmd, value, ch: 20, 16384, 2
cmd, value, ch: 22, 6, 2
cmd, value, ch: 17, 5, 2
cmd, value, ch: 18, 7, 2
cmd, value, ch: 39, 1500, 2
cmd, value, ch: 19, 1, 2
cmd, value, ch: 23, 47, 2
cmd, value, ch: 48, 1008981760, 2
cmd, value, ch: 49, 1017370373, 2
cmd, value, ch: 50, 1022739082, 2
cmd, value, ch: 40, -1057593754, 2
cmd, value, ch: 41, 1113876070, 2
cmd, value, ch: 42, -1199026020, 2
cmd, value, ch: 44, -1240050968, 2
cmd, value, ch: 46, 944076740, 2
cmd, value, ch: 43, -1158987046, 2
cmd, value, ch: 45, -1163117431, 2
cmd, value, ch: 47, -1148755919, 2
cmd, value, ch: 101, 5, 2
CH: 2
MCU_G-AHRS_5.21,FPGA-GP-PLL100-30
```

MCU cmd		FPGA cmd
102	CMD_DUMP_PARAMETERS	102
8	CMD_FOG_MOD_FREQ	0
13	CMD_FOG_WAIT_CNT	5
15	CMD_FOG_ERR_AVG	7
9	CMD_FOG_MOD_AMP_H	1
10	CMD_FOG_MOD_AMP_L	2
12	CMD_FOG_POLARITY	4
20	CMD_FOG_CONST_STEP	12
22	CMD_FOG_FPGA_R	14
17	CMD_FOG_GAIN1	9
18	CMD_FOG_GAIN2	10
39	CMD_FOG_CUTOFF	28
19	CMD_FOG_FB_ON	11
23	CMD_FOG_DAC_GAIN	50

48	CMD_THRESHOLD_WX	
49	CMD_THRESHOLD_WY	
50	CMD_THRESHOLD_WZ	
40	CMD_BIAS_COMP_T1	29
41	CMD_BIAS_COMP_T2	30
42	CMD_SFB_1_SLOPE	31
44	CMD_SFB_2_SLOPE	33
46	CMD_SFB_3_SLOPE	35
43	CMD_SFB_1_OFFSET	32
45	CMD_SFB_2_OFFSET	34
47	CMD_SFB_3_OFFSET	36
101	CMD_FPGA_VERSION	101

{"FREQ":99, "MOD_H":8842, "MOD_L":-8842, "ERR_OFFSET":0, "POLARITY":1, "WAIT_CNT":0, "ERR_TH":0, "ERR_AVG":0, "GAIN1":5, "GAIN2":7, "FB_ON":1, "CONST_STEP":16384, "HD_Q":1, "HD_R":6, "SF0":0.0000000000, "SF1":0.0005900000, "SF2":0.0000000000, "SF3":0.0000000000, "SF4":0.0000000000, "SF5":0.0000000000, "SF6":0.0000000000, "SF7":0.0000000000, "SF8":0.0000000000, "SF9":0.0000000000, "TMIN":-20, "T1":-10, "T2":0, "T3":10, "T4":20, "T5":30, "T6":40, "T7":50, "TMAX":80, "DAC_GAIN":47, "DATA_RATE":2000, "SFB":0.0000000000, "CUTOFF":1500, "BIAS_COMP_T1":-7.6999998093, "BIAS_COMP_T2":57.0999984741, "SFB_1_SLOPE":-0.0000650000, "SFB_1_OFFSET":-0.0017949000, "SFB_2_SLOPE":-0.0000022400, "SFB_2_OFFSET":-0.0013140599, "SFB_3_SLOPE":0.0000470800, "SFB_3_OFFSET":-0.0041313400,"STD_Wx":0.00999999,"STD_Wy":0.01999999,"STD_Wz":0.02999999}

按下校正 dump:

129: CMD_DUMP_CALI_PARAMETERS

```
cmd, value, ch: 129, 0, 2
```

DUMP VERSION:

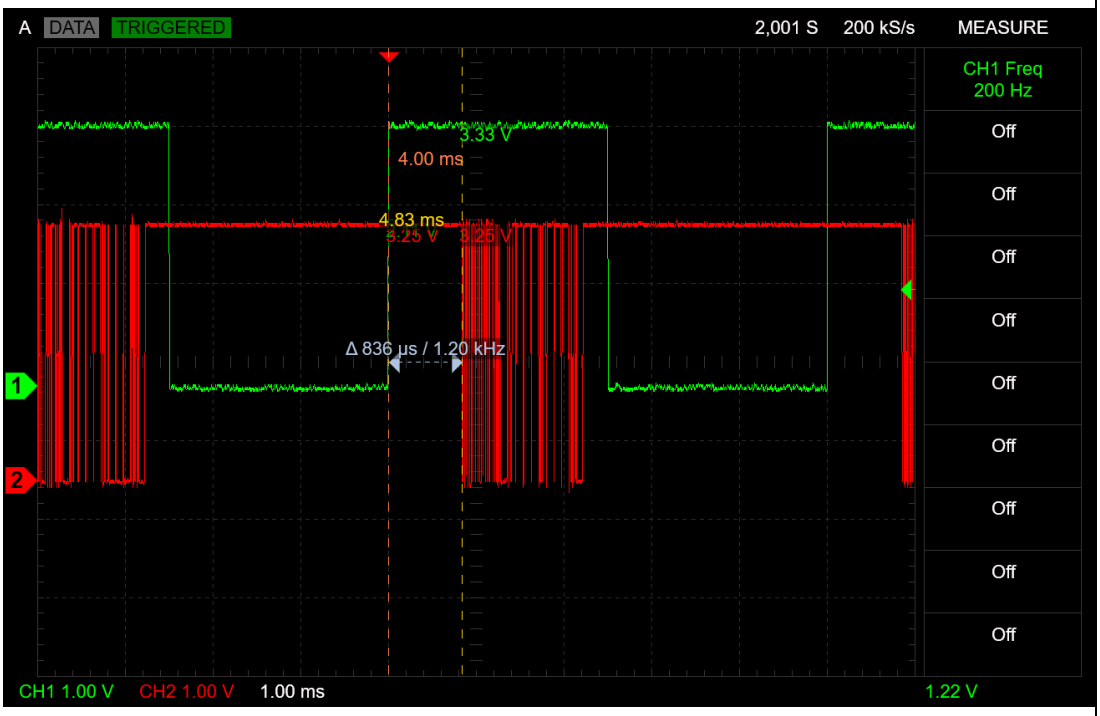
原本: MCU_G-AHRS_5.21,FPGA-GP-PLL100-30

舊時序:

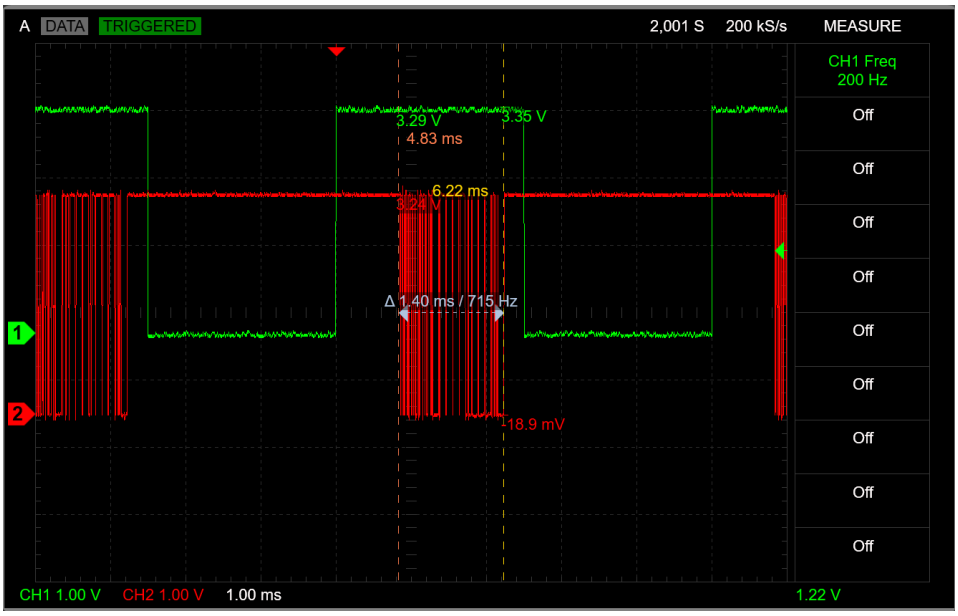
新時序:

SYNC vs FPGA Tx

200Hz: SYNC 到 FPGA Tx 0.84 ms

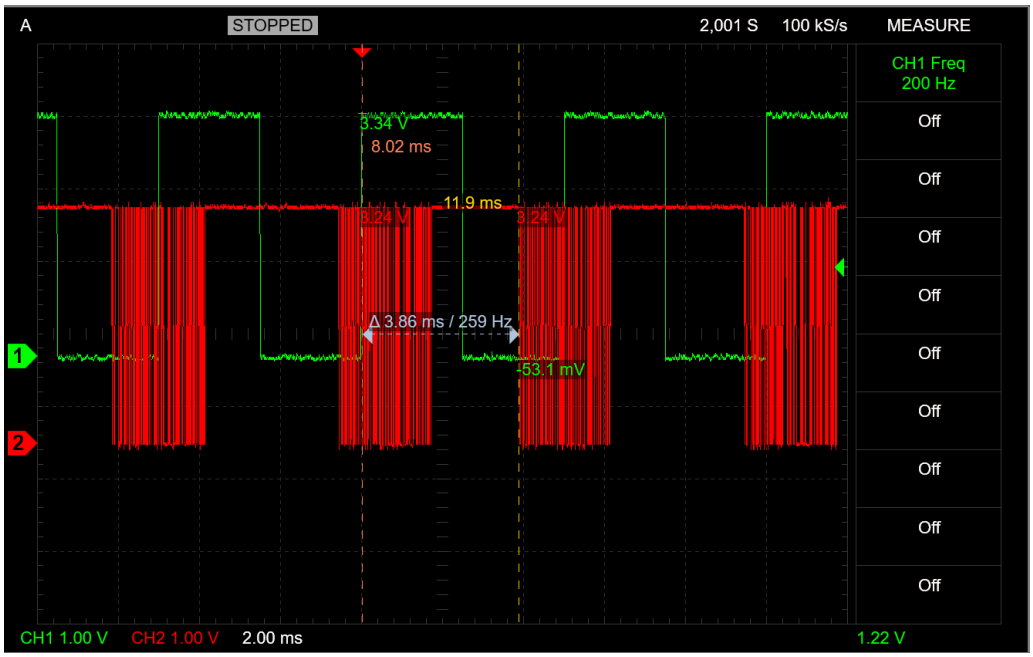


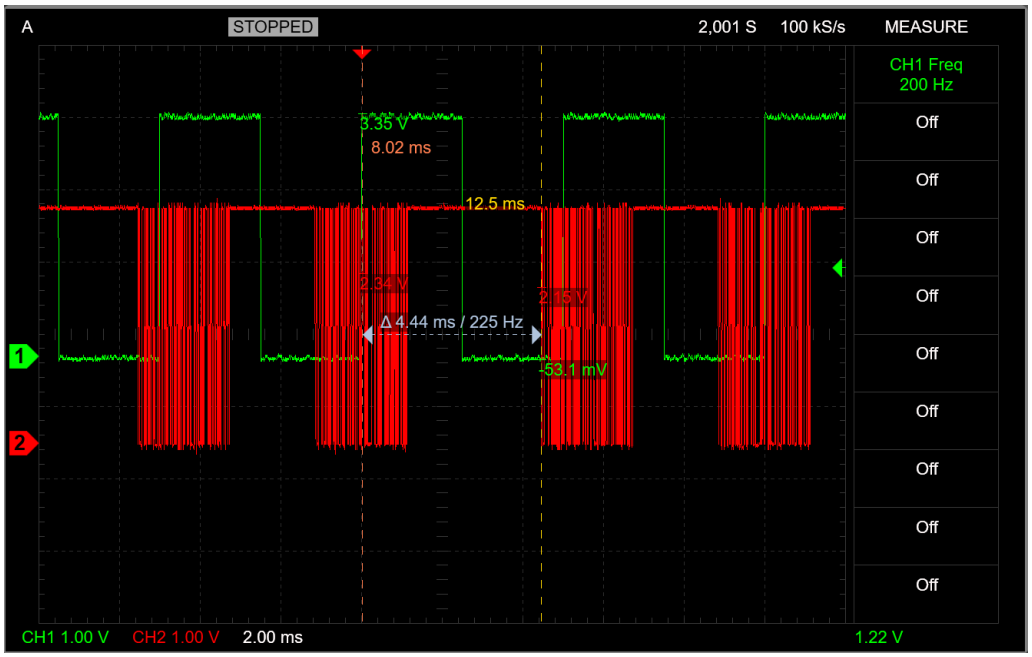
資料寬度 1.4 ms: 16 bytes = 160/115200 = 1.39 ms



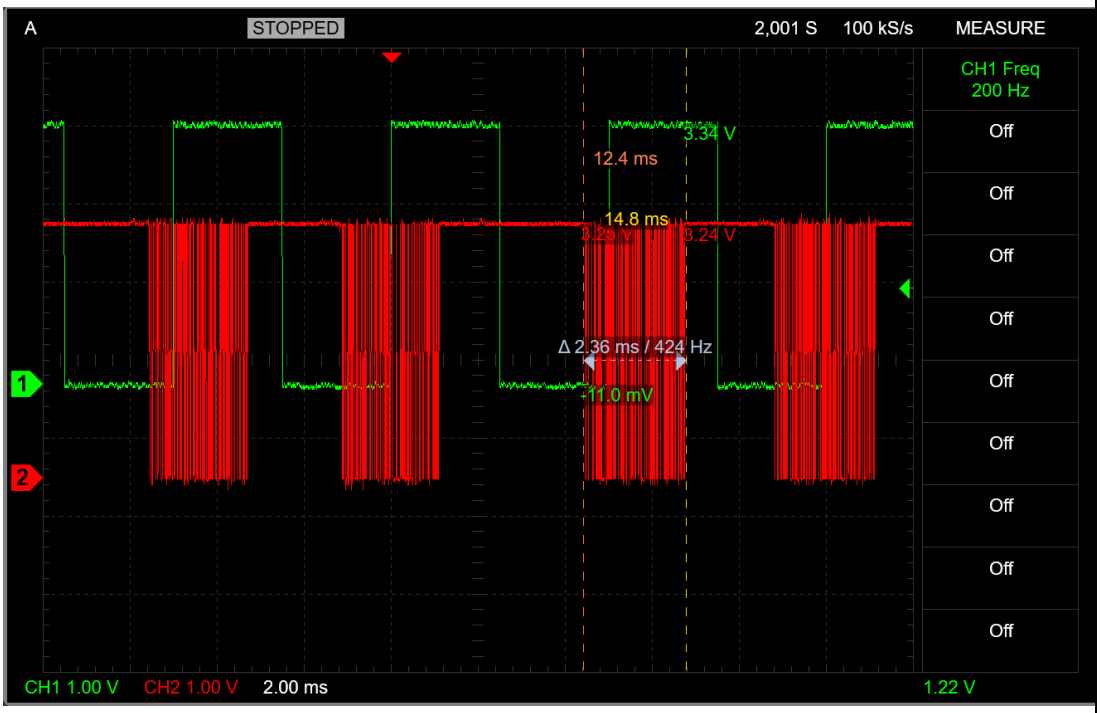
SYNC vs MCU Tx

3.86 or 4.44 ms，兩個狀態，應該是狀態機切成兩個 step

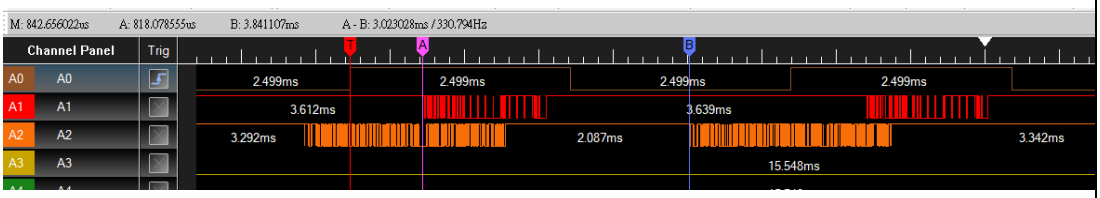




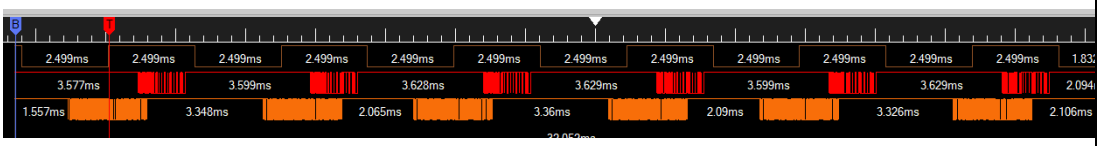
資料寬度 2.36 ms: 53 bytes = 530/230400 = 2.3 ms



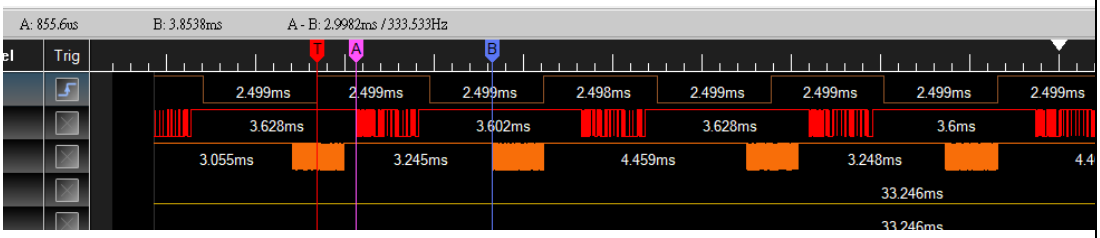
時序比較:



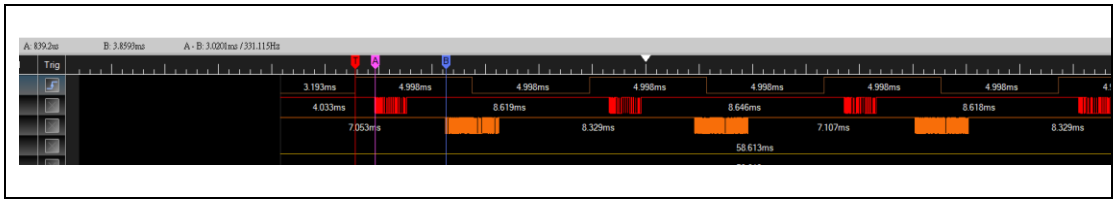
可看到 MCU Tx 間有兩種時間。 MCU Tx 跟 FPGA Tx 有時間重疊，若 Serial 中斷會造成問題嗎? 或會的話需要把 MCU Tx 改成 460800



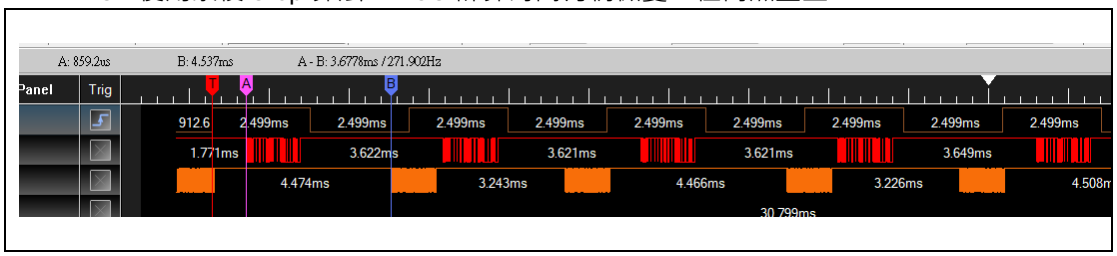
改成 460800，已不會重疊



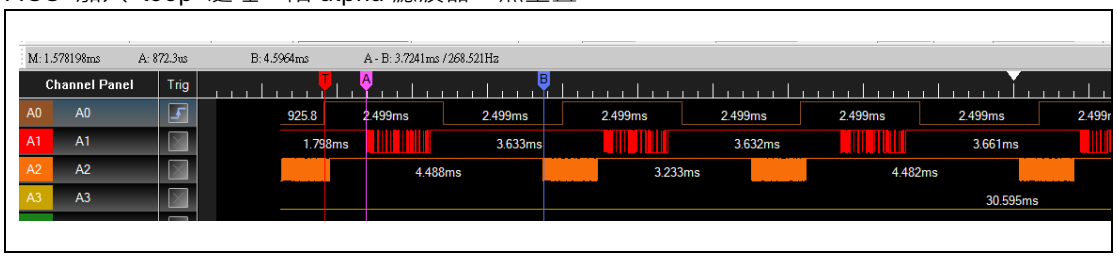
改成 100Hz，delay 時間相同，確定不是跨了一個週期



FPGA 使用累積 Step 算法，MCU 計算時間有稍微變，但尚無重疊



MCU 加入 loop 處理一階 alpha 濾波器，無重疊



繼續降頻寬:

目前 FPGA 以 200Hz 當作 sync，使用累積 step 方法讓輸出 BW 為 100Hz，因此 MCU 收到後計算得出的轉速值 BW = 100Hz。若想要再降低頻寬，可在 MCU 端執行 MV，其 window size 與 BW 關係:

下面 N=1 時為原始 BW

2. 不同 Window (N) 對應的頻寬計算

假設您的 $f_s = 200\text{Hz}$，以下是不同窗口長度對應的有效頻寬：

Window (N)	計算過程	預估頻寬 (f_{BW})	第一零點 (Null)
$N = 1$	$200/2$ (Nyquist)	**100 Hz**	無 (即原始數據)
$N = 2$	$0.443 \times 200/\sqrt{3}$	**~51.2 Hz**	**100 Hz**
$N = 4$	$0.443 \times 200/3.87$	**~22.9 Hz**	**50 Hz**
$N = 8$	$0.443 \times 200/7.93$	**~11.2 Hz**	**25 Hz**